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Effects of type traits, inbreeding, and production on survival in US Jersey cattle

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ABSTRACT

Greater longevity is associated with lower replacement costs, higher average milk production, and fewer replacement heifers needed. Longevity data are obtained late in life, and for this reason stayability, defined as the probability of survival from birth until a certain age, can be used as an alternative measure. The objective of this study was to evaluate the effects of different type traits, inbreeding, and production level on the stayability of Jersey cows to various ages, and to assess trends over time. Data consisted of 460,172 to 204,658 stayability records, depending on length of the opportunity period, for survival from birth until 36, 48, 60, 72, or 84 mo of age. Threshold models were used to analyze the stayability traits, including different type traits, inbreeding coefficient, and within-herd production level as explanatory variables. Heritability estimates for stayability traits ranged from 0.05 (36 mo) to 0.22 (84 mo). As expected, the probability of survival decreased as age increased. Highly productive cows were more likely to survive than their poor-producing contemporaries regardless of age and the type trait evaluated. Our data indicate that farmers' selection decisions tend to punish poor production at early ages and reward high production at later stages. Inbreeding negatively affected the probability of survival, especially when inbreeding coefficients exceeded 10%, and this impact was most noticeable at 48 mo of age or later. Some type traits, such as stature and foot angle, had little effect on the probability of survival. Other type traits, such as strength, dairy form, rump width, and rear legs, showed higher probability of survival at intermediate scores, whereas other type traits, such as fore udder attachment, rear udder height, udder depth, and final score, showed higher probability of survival at higher scores. Finally, our results indicate that the probability of survival has decreased in the last decade, probably due to a greater number of heifers available and, therefore, higher culling rates.

Key words: culling, probability of survival, stayability, threshold model

INTRODUCTION

Longevity, also known as length of productive life, is the period between a dairy cow's first calving and removal from the herd. Longevity is economically relevant because greater longevity is associated with lower replacement costs, higher average milk production, and fewer replacement heifers needed (Veerkamp and van Pelt, 2019). However, longevity cannot be recorded until late in life, and for this reason stayability, defined as the probability of survival from birth until a certain age, given the opportunity to reach that age, can be used as an alternative measure (Hudson and van Vleck, 1981).

Both voluntary and involuntary culling affect dairy cow longevity. Voluntary culling occurs when the farmer chooses to remove a cow from the herd due to poor production, whereas involuntary culling occurs when productive cows must be removed due to injuries, illness, poor reproduction, or death (Weigel et al., 2003; Fetrow et al., 2006). In general, voluntary culling due to low production involves an economic decision in which a new replacement heifer is expected to be more profitable than an existing cow, even after considering costs associated with the replacement (Fetrow et al., 2006; de Vries, 2017; de Vries and Marcondes, 2020). Although classifying reasons for culling into the distinct groups of voluntary and involuntary is common in the literature, more than one factor usually contributes to a culling decision in practice (Bascom and Young, 1998; Fetrow et al., 2006; Veerkamp and van Pelt, 2019).

The most common reasons for culling in Jersey cattle are low production, death, mastitis (including high SCC), and infertility (Norman et al., 2022). Type traits can influence longevity indirectly, because some measures of physical conformation are phenotypically or genetically correlated with various health disorders (Kern et al., 2015). Type traits related to the udder,

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such as udder depth, fore udder attachment, and teat placement, tend to have the most significant influence on longevity through their relationships with udder health, whereas feet and leg traits tend to have a modest influence through their effects on locomotion (Caraviello et al., 2003). Inbreeding can also affect longevity due to increased homozygosity and, consequently, the effects of inbreeding depression on health and performance (Bourdon, 2000). In Jersey cattle, inbreeding coefficients above 3% have been associated with increased risk of culling, and this risk increases at higher levels of inbreeding (Caraviello et al., 2003; Sewalem et al., 2006).

The objectives of this study were to describe the relationships of physical conformation and inbreeding with stayability of US Jersey cows to various ages, as well as their interactions with production level and the trends that have occurred over time due to changes in herd management practices and demographics.

MATERIALS AND METHODS

Data Set

Data consisted of 765,892 production and type records from Jersey cows that calved from 1994 to 2021. Data were provided by the American Jersey Cattle Association (**AJCA**; Reynoldsburg, OH). Only cows that have at least 4 generations of known Jersey ancestors (Jersey level equal to 4 or higher), with age at first calving between 550 and 1,000 d, and at least 1 test day record were considered in the analyses. No animal subjects were used, so this analysis did not require approval by an Institutional Animal Care and Use Committee.

Response Variable

Stayability was evaluated as survival from birth until 36 (**STAY36**), 48 (**STAY48**), 60 (**STAY60**), 72 (**STAY72**), and 84 mo of age (**STAY84**). Cows born after January 1 of 2018, 2017, 2016, 2015, and 2014 were removed from the STAY36, STAY48, STAY60, STAY72, and STAY84 data sets, respectively, to give all animals an opportunity to fully express the phenotypes. Each cow in the data set had a termination code indicating status of the animal at the end of the most recent recorded lactation, and age at culling was considered as the age when the last recorded lactation ended.

Stayability was considered as a success for animals that survived until the next herd test day after reaching a given age. Cows that did not reach that age and that had termination codes indicating death or sold for slaughter or salvage were considered as failures. Cows that did not reach the age and that had termination codes indicating they were still in the herd or had been sold for dairy purposes were considered as censored and removed from the analysis, because it was not possible to determine age at culling.

Statistical Model

A probit model (Gianola and Foulley, 1983) was used to analyze the stayability traits. This model describes an observed response variable (Y) using an underlying linear model,

$$z = \boldsymbol{\eta} + \boldsymbol{\varepsilon},$$

where z is the underlying variable (liability), $\boldsymbol{\eta}$ is a vector of linear predictors, and $\boldsymbol{\varepsilon}$ is a vector of independent and identically distributed standard normal random variables.

The observable outcome is $Y = \{1 \text{ if } z > 0;\ 0 \text{ otherwise}\}$, indicating that $Y = 1$ when the liability is greater than zero. Therefore, the conditional probability of survival is

$$P(Y = 1 \mid \boldsymbol{\eta}) = \Phi(\boldsymbol{\eta}),$$

where $\Phi(.)$ is the standard normal cumulative distribution function. Hence, the likelihood function becomes

$$p(\boldsymbol{Y} \mid \boldsymbol{\eta}) = \Pi\Phi(\boldsymbol{\eta})^{\boldsymbol{Y}}\cdot[1 - \Phi(\boldsymbol{\eta})]^{1-\boldsymbol{Y}},$$

where $\boldsymbol{Y}$ and $\boldsymbol{\eta}$ are vectors of response variables and the linear predictors, respectively.

For stayability, the following linear predictor was considered:

$$\boldsymbol{\eta} = \boldsymbol{HYS} + \boldsymbol{JX} + \boldsymbol{PROD} + \beta_1\boldsymbol{AFC} + \beta_2\boldsymbol{AFC}^2 + \beta_3\boldsymbol{F} + \beta_4\boldsymbol{F}^2 + \boldsymbol{\beta}_5\boldsymbol{T}(\boldsymbol{PROD}) + \boldsymbol{\beta}_6\boldsymbol{T}^2(\boldsymbol{PROD}) + \boldsymbol{a},$$

where $\boldsymbol{HYS}$ is the fixed effect of contemporary group; $\boldsymbol{JX}$ is the fixed effect of Jersey level (4 levels); $\boldsymbol{PROD}$ is the fixed effect of within-contemporary group production level (4 levels); $\boldsymbol{AFC}$ is the fixed effect of age at first calving, considered as linear and quadratic; $\boldsymbol{F}$ is the fixed effect of inbreeding coefficient, considered as linear and quadratic; $\boldsymbol{T}(\boldsymbol{PROD})$ is the fixed effect of standardized type trait, considered as linear and quadratic, nested within production level (see below); $\boldsymbol{a}$ is a vector of random additive genetic effects; β_1, β_2, β_3, and β_4 are the corresponding regression coefficients; and $\boldsymbol{\beta}_5$ and $\boldsymbol{\beta}_6$ are the corresponding vectors of regression coefficients.

Contemporary groups were formed by the combination of herd, year, and season of first calving. Seasons were defined as winter (December to February), spring (March to May), summer (June to August), and fall (September to November). Contemporary groups with fewer than 10 animals were removed from the analysis. Connectedness among contemporary groups was tested using AMC software (Roso and Schenkel, 2006). Contemporary groups with fewer than 10 direct genetic links and those comprised of daughters of a single sire were discarded.

Production level was evaluated as a categorical trait with 4 levels: poor, low, medium, and high. Those 4 levels correspond to the quartiles of the sum of the standardized mature-equivalent 305-d fat and protein production of the last lactation recorded within contemporary group.

The functional type traits evaluated were final score, stature, strength, dairy form, rump angle, rump width, rear legs, fore udder attachment, rear udder height, rear udder width, foot angle, udder depth, udder cleft, front teat placement, front teat length, tear teat placement (side view), and rear teat placement (rear view). A complete description of each trait is available at AJCA (2022). Only type appraisals during the first lactation were considered, because appraisal data during the second and later lactations were sparse. For cows with more than one appraisal during first lactation, we considered the appraisal taken closest to 150 DIM. Final score ranged from 50 to 89 points because 89 is the maximum value allowed for primiparous cows. For stature, scores ranged from 1 to 80, whereas scores for the remaining traits ranged from 1 to 50 (AJCA, 2022). Type scores were standardized for age and stage of lactation using the residuals of the linear model that considered the linear and quadratic effects of age at first calving and days in milk at the time of appraisal.

The animal effect was assumed to follow the following normal distribution,

$$\left(\boldsymbol{a} \mid \sigma_a^2\right) \sim N\left(0, \boldsymbol{A}\sigma_a^2\right),$$

where $\boldsymbol{a}$ is the vector of additive genetic effect, σ_a^2 is the additive genetic variance, and $\boldsymbol{A}$ is the matrix of additive relationships between animals in the pedigree file. Animal and residual effects were assumed to be independent of each other.

Implementation

Data editing and calculation of summary statistics were performed using R software (version 4.2.1; R Core Team, 2022). A univariate threshold model was implemented for stayability to each age in a Bayesian framework using the software THRGIBBS1F90 (version 2.110; Misztal et al., 2022). The effect of each type trait was included one at a time in the model due to strong correlations between some pairs of traits (Caraviello et al., 2003). Inferences were made based on 80,000 samples after discarding the first 20,000 samples as burn-in. A thinning interval of 10 was used to compute statistics of the posterior distribution using POSTGIBBSF90 software (version 3.08; Misztal et al., 2022).

Heritability Estimation

The heritability estimate (h^2) was calculated using the following standard formula:

$$h^2 = \frac{\sigma_a^2}{\sigma_a^2 + \sigma_e^2},$$

where σ_a^2 is the additive genetic variance and $\sigma_e^2 = 1$.

Predictions

The conditional expectation function of the probit model, evaluated at the posterior mean of model unknowns, is $y = \Phi(\boldsymbol{\eta})$, where Φ is the standard normal cumulative distribution function. To study the effects of production level, inbreeding, and type traits on stayability, we considered the mean age at first calving (703 d), purebred animals, and the largest contemporary group with average stayability near the population observed average. To evaluate the importance of inbreeding, type trait deviations were considered as the population mean (zero, after standardization), and to evaluate the importance of type traits, the average inbreeding coefficient (0.07) was assumed. The effect of time was evaluated by comparing 2 different calving periods: 2000 to 2005 and 2010 to 2015, and by computing estimated stayability by year of calving over the entire data collection period.

RESULTS

The numbers of cows with phenotypes, sires, dams, and contemporary groups for each age are presented in Table 1. The size of the data set ranged from 460,172 for STAY36 to 204,658 for STAY84. Fewer cows remained in the data set as the age increased, because cows born in the most recent years did not yet have an opportunity to express the stayability phenotype. Inbreeding ranged from 0 to 42.5%, and almost 90% of cows with phenotypes had inbreeding coefficients between 0 and 10%.

Table 1. Summary of the data sets used in threshold model analysis of Jersey cow stayability to different ages

Data set	36 mo	48 mo	60 mo	72 mo	84 mo
Cows with phenotypes (n)	460,172	369,874	298,521	243,946	204,658
Sires (n)	9,689	8,849	8,116	7,423	6,756
Dams (n)	336,916	278,724	231,413	193,868	165,713
Contemporary groups (n)	14,114	13,488	12,749	12,060	11,302

Heritability Estimates

The h^2 estimates for stayability traits (±SE) varied between 0.05 ± 0.004 and 0.06 ± 0.005 for STAY36, depending on which type trait was considered as a covariate, from 0.07 ± 0.005 to 0.08 ± 0.007 for STAY48, from 0.11 ± 0.007 to 0.13 ± 0.008 for STAY60, from 0.15 ± 0.009 to 0.18 ± 0.009 for STAY72, and from 0.18 ± 0.024 to 0.22 ± 0.013 for STAY84 (Supplemental Table S1, https://data.mendeley.com/datasets/sz4kkrcbw5/3; Mazetti Nascimento, 2023). Both h^2 estimates and their standard errors increased as the age increased, because animals had greater opportunity to express their genetic potential, and thus greater genetic variation in stayability, with advancing age. Standard errors increase as well, because fewer animals remained in the analysis at higher ages.

Effects of Inbreeding and Production Level on Stayability

As expected, the probability of survival decreased as the age increased (Figure 1). Highly productive cows were more likely to survive than poor-producing cows, regardless of age. Note in Figure 1 that the effect of production on the probability of survival was less critical for STAY36 compared with other ages, and differences between production levels in probability of survival were greatest for STAY60 and STAY72. Interestingly, cows in the lowest production quartile deviated most noticeably from the other groups for STAY36 and STAY48, whereas cows in the highest quartile had the largest deviations from other production groups for STAY72 and STAY84 (Figure 2). This seems to indicate that farmers tend to punish poor production at early ages and reward high production at later stages when making culling decisions. Moreover, inbreeding negatively affected the survival probability, especially when inbreeding coefficient exceeded 10%. This effect of inbreeding on survival was most noticeable for STAY48, STAY60, and STAY72 (Figure 2). For instance, the probability of survival decreases at a rate of 0.25% for STAY60 at a medium production level when the inbreeding coefficient increases from zero to 10%. When the inbreeding coefficient increases from 10% to 20%, the rate of decrease is 0.55%. The same trend was observed for the other production levels, as well as stayability to other ages, such as STAY48 and STAY72.

Effect of Type Traits on Stayability

Figures 3 to 7 show the effects of type traits on STAY60, the closest age to median survival in this population. For final score (Figure 3), the probability of STAY60 varied from 15 to 40% across production levels for cows with low type scores, whereas cows with high type scores had a 70 to 80% probability of survival across production levels. Differences in survival probability between production level were smaller as the final score increased, indicating possible favoritism for cows with high type scores and modest production. Estimated STAY60 probabilities were similar across type scores at each production level for stature (Figure 4). For this trait, the type score had little influence on the likelihood of the cow remaining in the herd, demonstrated by nearly flat curves, except for the lowest production level, where probability of survival was lower for cows with extremely high or low scores. For strength, dairy form, and rump angle (Figure 4), intermediate scores were associated with higher probabilities of survival to 60 mo of age. Optimum scores for strength and dairy form were slightly above average, with greatest impair-

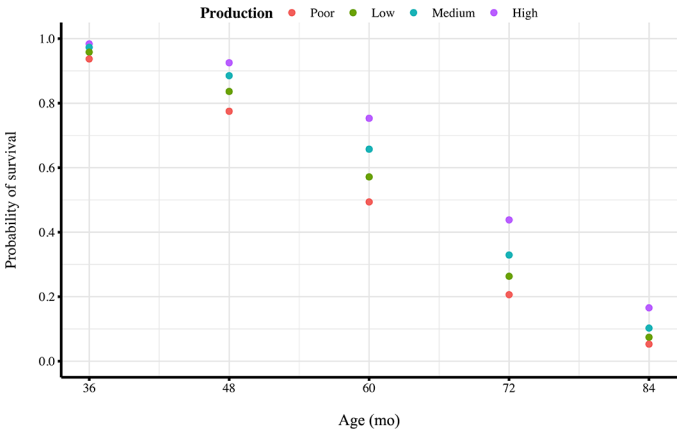


Figure 1. Effect of production level on the probability of survival at 36, 48, 60, 72, and 84 mo of age in Jersey cattle. This analysis considered final score as type trait.

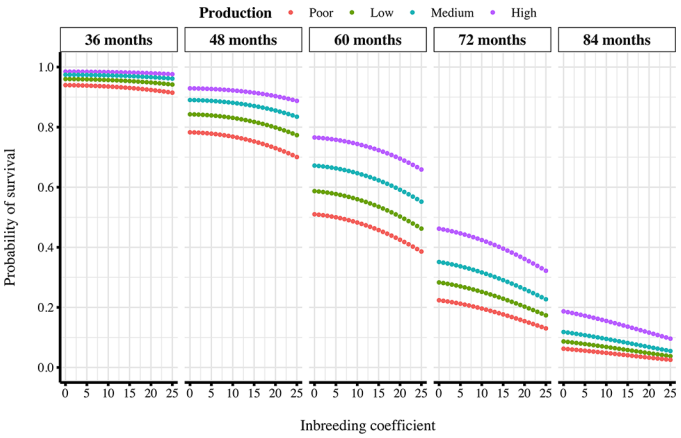


Figure 2. Effect of inbreeding by production level on the probability of survival at 36, 48, 60, 72, and 84 mo of age in Jersey cattle. This analysis considered final score as type trait.

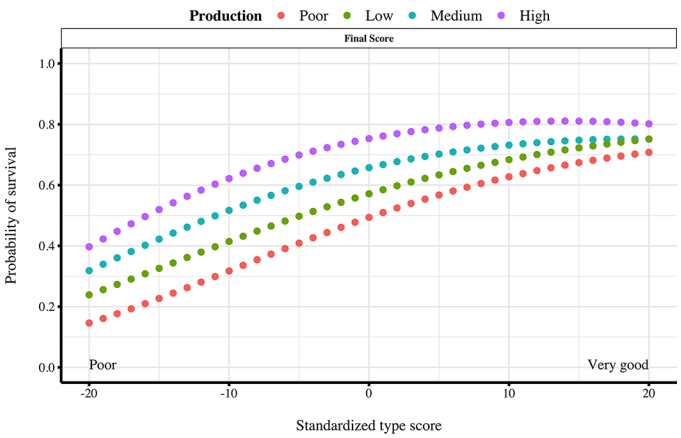


Figure 3. Effect of final score by production level on the probability of survival until 60 mo of age in Jersey cattle.

ment in STAY60 associated with poor-producing cows with low type scores. Conversely, optimum rump angle scores were slightly below average, with no difference in STAY60 between the extremes. The STAY60 pattern for rump width (Figure 5) was very similar to that of strength and dairy form, with an optimum slightly above breed average and poorest survival at the lower extreme, whereas for rear legs STAY60 was maximized for scores slightly below average and poorest survival for cows with extremely high scores (sickled rear legs). Increases in STAY60 probability were nearly linear, with increasing scores for fore udder attachment and rear udder height (Figure 5) at higher production levels, but an interaction was evident for lower production levels, where gains in STAY60 diminished as type scores increased.

Rear udder width (Figure 6) had minimal influence on STAY60 for high-producing cows, but low scores corresponding to narrow rear udder were increasingly detrimental to survival as production level decreased. Foot angle (Figure 6) had a modest relationship with survival, but a small advantage was observed in STAY60 in median scores in all production levels, except medium production level, where STAY60 increases toward steeper foot angle. Udder depth (Figure 6) had a large effect on STAY60, with impaired survival for cows with low scores and deep udders, regardless of production level. However, an interaction between production and type was observed as scores increased: the optimal probability of survival was above the mean score in high-producing cows, and closer to the mean as the production level decreased. The slope of STAY60 after the optimal values toward shallower udders was more substantial as the production level decreased. Graphs for udder cleft (Figure 6) and front teat placement (Figure 7) were nearly identical, with greatest STAY60 for cows with scores slightly above the population mean, impaired survival at both extremes, and poorest survival for cows with very low type scores. Lower scores for front teat length (Figure 7), corresponding to shorter teats, were slightly preferred among high-producing cows, but intermediate scores were associated with greatest STAY60 among poor-producing cows. Finally, rear teat placement (Figure 7) displayed an intermediate optimum for side view across all production levels, except low production level. For the rear view, STAY60 was greatest for cows with scores slightly above the mean (teats closer together).

Changes in Survival Trends Over Time

Many factors, such as management practices, facilities, equipment, and available biotechnologies, can change over time, and these can influence culling pressures or preferences. Figure 8 displays STAY60 probabilities for final score for cows that calved from 2000 to 2005 and cows that calved from 2010 to 2015. Cows with higher scores were more likely to survive in both time periods, but a clear interaction was observed. In the earlier time period, STAY60 varied widely between production levels for cows with low type scores, but cows with the highest type score had a roughly 80% chance of survival regardless of production level. In contrast, production level had a consistent effect on STAY60 for cows that calved between 2010 and 2015, such that high-producing cows had a roughly 20% greater chance of survival than poor-producing cows, regardless of final score. When the probability of survival was evaluated by udder depth score during these 2 time periods (Figure 9), the behavior of the curves

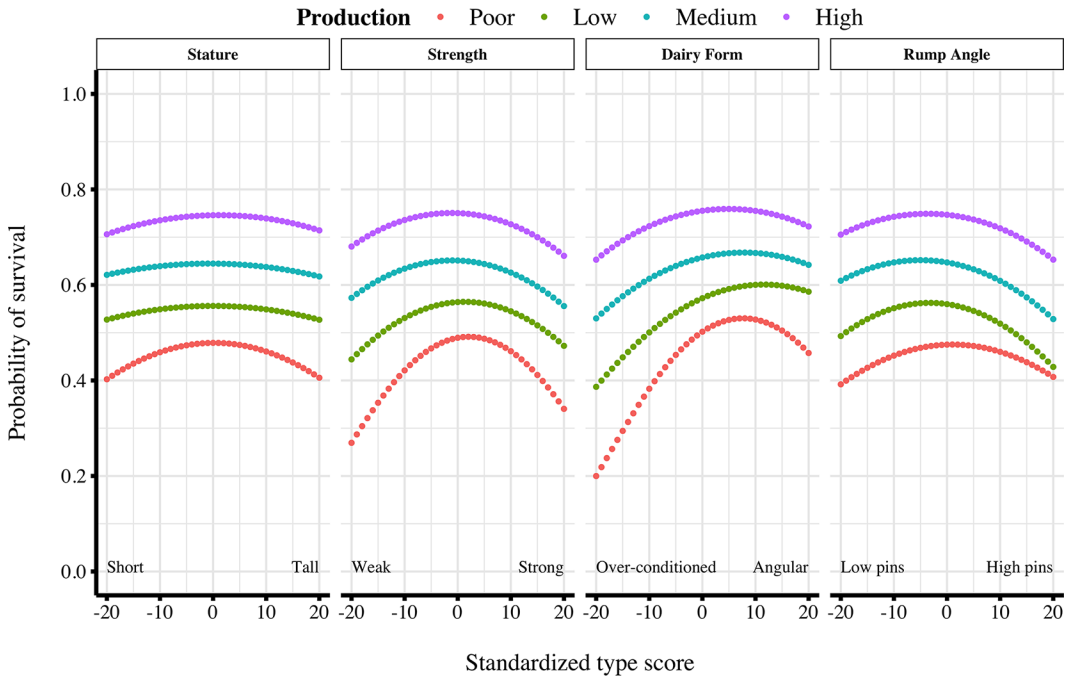


Figure 4. Effects of stature, strength, dairy form, and rump angle, by production level, on the probability of survival until 60 mo of age in Jersey cattle.

was similar to that of final score. Cows with deep udders had less chance to survive than cows with shallow udders in both periods and across production levels. Higher probabilities of survival were observed in type scores slightly above the mean in all production levels, except for high-productive cows in the earlier period.

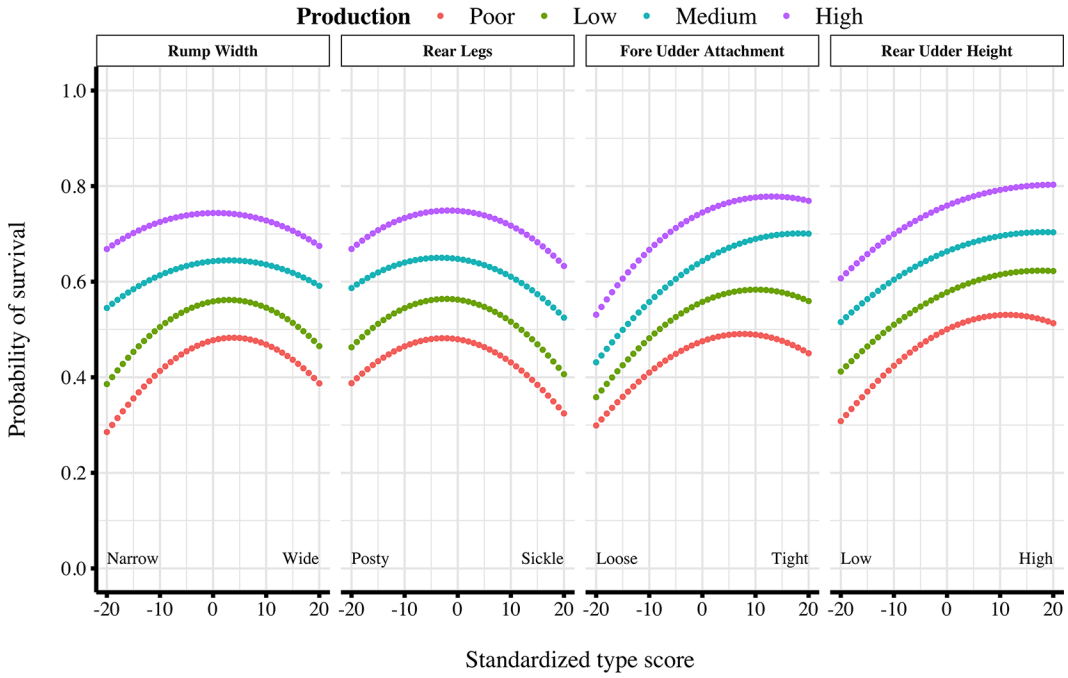


Figure 5. Effects of rump width, rear legs, fore udder attachment, and rear udder height, by production level, on the probability of survival until 60 mo of age in Jersey cattle.

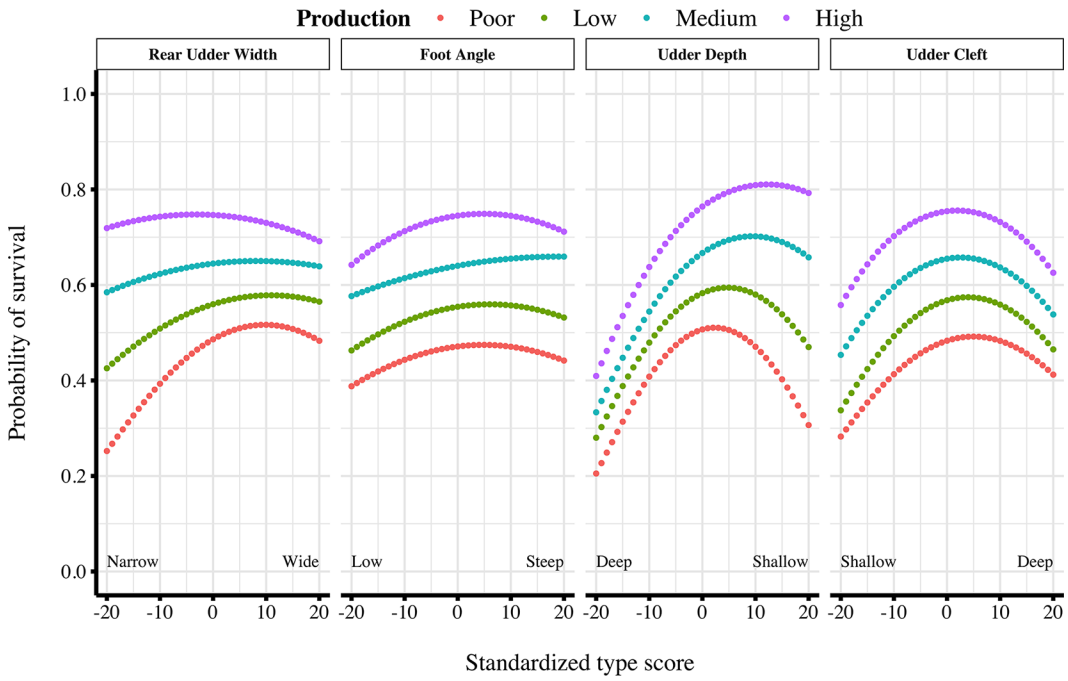


Figure 6. Effects of rear udder width, foot angle, udder depth, and udder cleft, by production level, on the probability of survival until 60 mo of age in Jersey cattle.

For those animals, shallower udders were associated with higher survival probability. A similar trend was observed for the later period, but the optimal type trait scores for survival for low- and medium-producing cows were closest to high scores in this period. Production level had less influence on probability of survival for

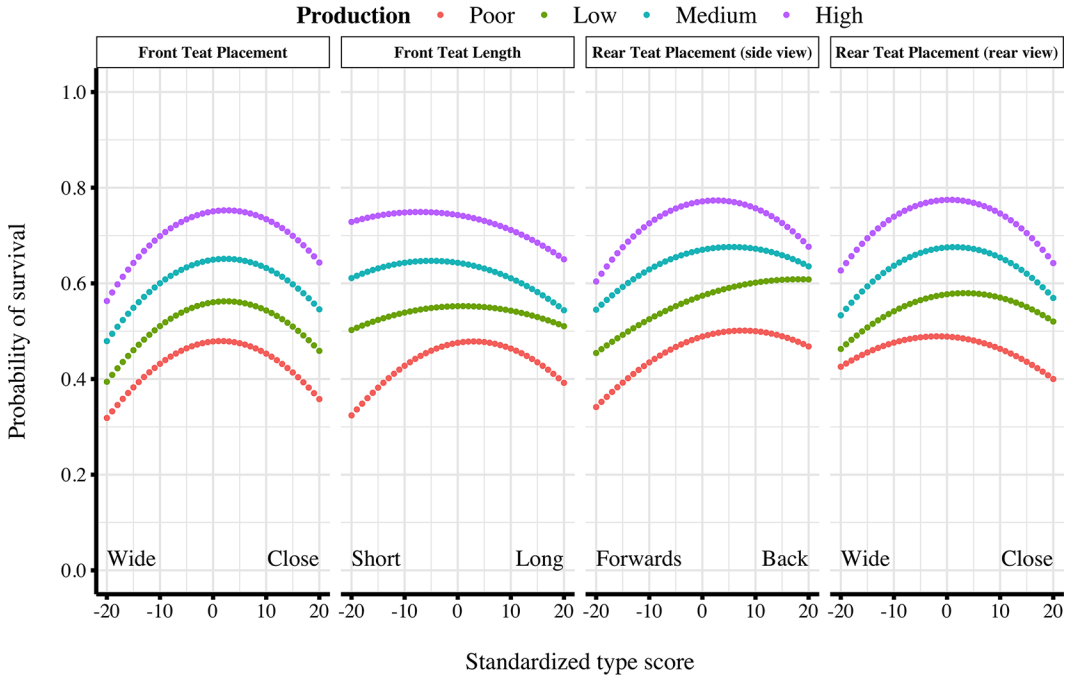


Figure 7. Effects of front teat placement, front teat length, rear teat placement (side view), and rear teat placement (rear view), by production level, on the probability of survival until 60 mo of age in Jersey cattle.

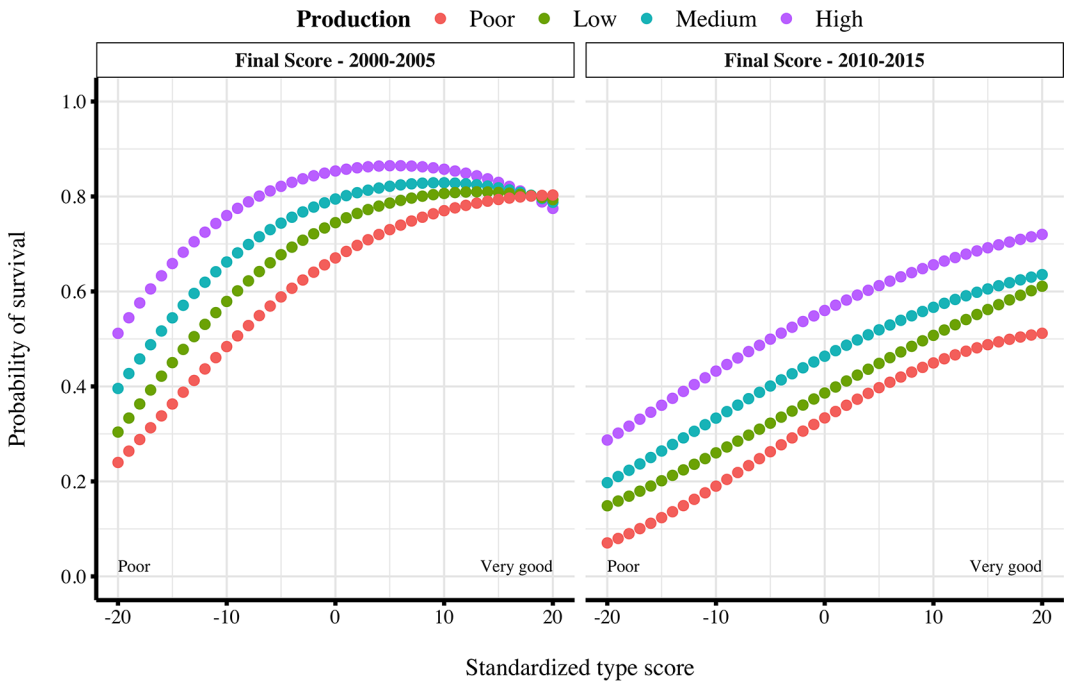


Figure 8. Effect of final score, by production level, on the probability of survival until 60 mo of age for 2 different calving periods in Jersey cows.

cows with low udder depth scores (i.e., with deep udders), especially in the later time period.

Figure 10 shows the trend in STAY60 for cows in the medium production quartile across all calving years in the study. Probability of survival to 60 mo of age was nearly constant from 2000 to 2007; after that, a significant decrease in STAY60 was observed, indicating greater culling pressure in recent years. Note that dots

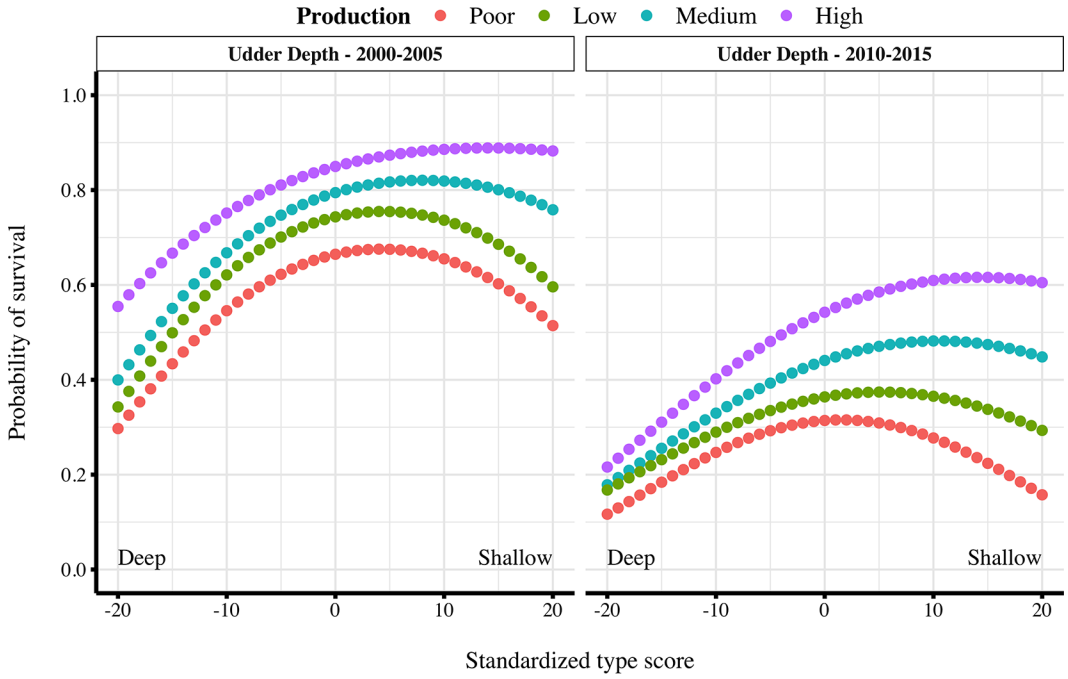


Figure 9. Effect of udder depth, by production level, on the probability of survival until 60 mo for 2 different calving periods in Jersey cows.

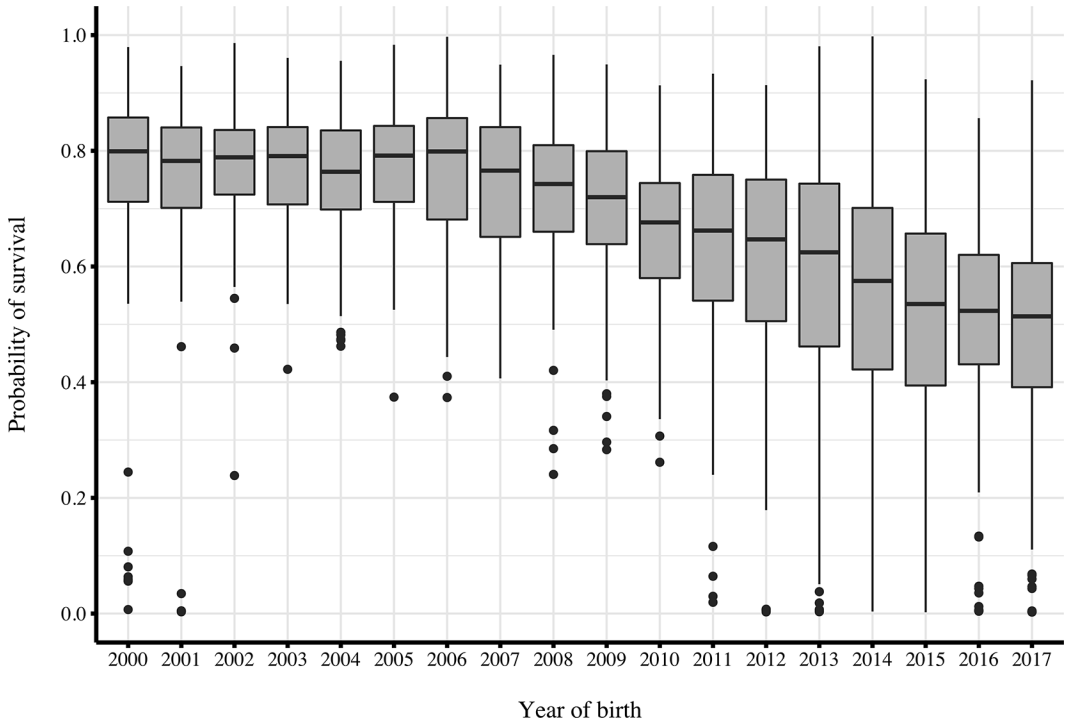


Figure 10. Probability of survival until 60 mo of age for medium-producing Jersey cows by year of calving. This analysis considered final score as type trait. The bottom and top of the box represent first and third quartiles; the vertical line denotes the median; the whiskers correspond to 1.5 × interquartile distance; and dark dots are outliers.

corresponding to extremely low survival probabilities represent outlier contemporary groups in which most or all animals were culled, perhaps due to farms exiting the industry.

Correlations among breeding values for stayability to different ages were high, varying from 0.44 between STAY36 and STAY84, to 0.86 between STAY72 and STAY84. As expected, the closer ages showed higher correlations between them than more distant ages.

DISCUSSION

Greater longevity in dairy cattle is associated with reduced replacement costs and increased average milk production. Farmers voluntarily cull low-producing cows for economic reasons, with the aim of increasing farm profitability. However, poor reproduction, injuries, and illnesses can force the involuntary culling of productive animals, which tends to increase veterinary costs and decrease farm profitability. In addition to milk yield, linear type traits and inbreeding can be related to culling, either by direct effects on productivity or by indirect associations with incidences of illness, injury, or infertility. Because longevity is measured very late in life, stayability can be used as an alternative measure. In our study, modest heritability estimates for STAY36, STAY48, STAY60, STAY72, and STAY84 suggest that selection for improved stayability would lead to slow genetic gains. Similar results were found by Kern et al. (2014), who evaluated the longevity of Holstein cattle with a threshold model. In the lifetime net merit index (NM$) published by the Council on Dairy Cattle Breeding (Cole et al., 2021), productive life is used to measure the time from first calving until culling, and it receives a relative economic emphasis of nearly 20% in Jersey cattle (VanRaden et al., 2021). Hence, despite the potential for slow genetic progress due to low heritability, selection for longevity is of major importance in US dairy cattle. Additionally, stayability until earlier ages can be an appropriate selection criterion for longevity, because it is highly and favorably correlated with survival until older ages.

Type traits are indirectly related to longevity, through their association with the incidence of various diseases and injuries. For example, poor conformation of feet and legs has been reported to be genetically associated with increased lameness (Khansefid et al., 2021). However, the relationship between probability of survival and foot angle and rear legs scores in Jersey and Holstein-Friesian cows in previous studies has been small (Caraviello et al., 2003; Williams et al., 2022a). Our study found a modest relationship between

STAY60 and foot angle score, but rear leg score seemed to have stronger effects on survival, with impaired survival at both extremes. Rump angle scores reflecting low or high pins may be related to calving difficulties (Sewalem et al., 2004), which can increase the probability of culling, and our results showed that cows with intermediate rump angle were more likely to survive to 60 mo of age. Poor teat placement can be related to lower machine milking efficiency and higher incidence of subclinical mastitis (Singh et al., 2017), and our study demonstrated that cows with more central teat placement, and therefore intermediate scores for front teat placement and rear teat placement (side and rear views), had greater probability of survival. Other type traits related to the udder, such as udder depth, udder cleft, fore udder attachment, rear udder height, and rear udder width, are associated with differences in milk production and susceptibility to injuries. Caraviello et al. (2003) reported that udder traits were the most important indicators of longevity in Jersey cattle, among all type traits considered. The authors observed a high risk of culling for animals with low scores for those traits, similar to the present study, and often with diminishing returns in survival probability as type scores approached their maximum. As we observed in our study, stature was the least important trait with respect to survival in previous studies with Holstein and Jersey cattle (Caraviello et al., 2003; Sewalem et al., 2004). Notably, within-herd ranking for milk production has a strong influence on culling decisions, and its relative importance versus type traits seems to have increased over time. In the past, cows with high final scores had similar probabilities of survival regardless of their level of milk production. But in recent years, it appears that cows with low milk production are culled from the herd regardless of their high type scores.

Inbreeding depression has important direct effects on performance due to increased homozygosity, which can lead to increased culling because of low production, longer calving intervals, lower conception rates, and higher somatic cell scores (Doekes et al., 2019). Caraviello et al. (2003) observed an increase in the risk of culling for animals with inbreeding coefficients above 7%, and, as in our study, they reported a nearly linear relationship between risk of culling and inbreeding coefficients. Sewalem et al. (2006) observed that Jersey cows with inbreeding coefficients exceeding 12.5 were roughly 30% more likely to be culled than their less-inbred contemporaries. Furthermore, the effect of inbreeding on survival may increase across parities, as du Toit et al. (2012) observed a stronger association between inbreeding and survival in second-lactation Jersey cattle in South Africa than in their first-parity contemporaries.

Genetic improvement of longevity can have a significant economic impact on dairy herds, because mature cows that are highly productive and demand few veterinary interventions are very profitable (García-Ruiz et al., 2016; Dallago et al., 2021). In addition, most cows do not repay their rearing costs until after the second lactation (Boulton et al., 2017), and involuntary culling of cows at an early age leads to economic losses for the farmer. Increases in longevity can also influence environmental sustainability and animal welfare. For example, an increase in annual herd replacement rates can contribute to higher greenhouse gas emissions at the whole-farm level (Wall et al., 2012; Grandl et al., 2019), and higher involuntary culling rates in young animals may indicate poor welfare, which can influence consumers' perceptions of and demand for dairy products. In addition, surplus heifers, if not needed as herd replacements, can be sold to provide an additional source of farm income (Dallago et al., 2021). Finally, by coupling the use of sexed semen on young cows and heifers with the use beef semen on older cows, farmers can reap the profits of high milk production from mature cows without compromising genetic progress, allowing mature cows that are healthy and reproductively sound to remain in the herd for 1 or 2 additional lactations.

Changes in management practices, facilities, herd size, economic pressures, and other factors have reduced the probability of cow survival in commercial dairy herds. For example, with the advent of sexed semen, a larger number of heifers have become available as herd replacements. In addition, in situations where facilities or available acreage limit herd size, farmers may be pushed to increase culling rates (de Vries, 2020). Furthermore, genomic selection has enhanced genetic progress and generated replacement heifers with increasingly high milk production potential (Meuwissen et al., 2001; Williams et al., 2022b). These factors, and others, have led to reduced probability of survival to advanced ages in Jerseys and other breeds, despite genetic improvements in productive life and other fitness traits. This apparent paradox results from ongoing changes in the balance of voluntary (economic) and involuntary (forced) culling, and breed improvement programs should seek to improve the genetic predisposition of animals to remain healthy, fertile, and productive, as long as they can make greater economic contributions to the herd than their potential replacements.

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